

Assignment 0 – Homogeneous Coordinate Transformation

Problem

Given a point:

$$P = (2, 1)$$

First rotate the point 45° counterclockwise around the origin, then translate it by (1, 2).

Calculate the final coordinate using homogeneous coordinates.

A 2D point (x, y) can be written as:

$$P = [x, y, 1]^T$$

So:

$$P = [2, 1, 1]^T$$

Assume the original point can be represented using polar coordinates:

$$x = r \cos\theta$$

$$y = r \sin\theta$$

After rotating the point counterclockwise by angle ϕ :

$$x' = r \cos(\theta + \phi)$$

$$y' = r \sin(\theta + \phi)$$

Using trigonometric identities:

$$\cos(\theta + \phi) = \cos\theta \cos\phi - \sin\theta \sin\phi$$

$$\sin(\theta + \phi) = \sin\theta \cos\phi + \cos\theta \sin\phi$$

Therefore:

$$x' = r(\cos\theta \cos\phi - \sin\theta \sin\phi)$$

$$y' = r(\sin\theta \cos\phi + \cos\theta \sin\phi)$$

$$x' = x \cos\phi - y \sin\phi$$

$$y' = x \sin\phi + y \cos\phi$$

So the rotation matrix is:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

For $\phi = 45^\circ$:

So after rotation:

$$P_{\text{rotated}} = [0.707, 2.121, 1]^T$$

Now apply the translation matrix to the rotated point:

$$\begin{bmatrix} 0.707 \\ 2.121 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1.707 \\ 4.121 \\ 1 \end{bmatrix}$$

$$P' = (1.707, 4.121, 1)$$